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AI PRACTICAL NO. 10

AIM: Demo of OpenAI/TensorFlow Tools

- Explore and experiment with OpenAI or TensorFlow tools and libraries.
- Perform a demonstration or mini-project showcasing the capabilities of the tools.
- Discuss and present the findings and potential applications.

#CODE:

```
# PRACTICAL 10: DEMO OF TENSORFLOW TOOLS
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf

from google.colab import files

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)
# Upload dataset
uploaded = files.upload()
# Read dataset
df = pd.read_csv("cities_map.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumns:")
print(df.columns)

# FEATURES AND TARGET

X = df[
    [
```

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```
        "Latitude",
        "Longitude"
    ]
]

y = df["Population_Million"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=42
)

# Scale input features
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(
    X_train
)

X_test_scaled = scaler.transform(
    X_test
)

# BUILD TENSORFLOW MODEL

model = tf.keras.Sequential([
    tf.keras.layers.Input(
        shape=(2,)
    ),

    tf.keras.layers.Dense(
        16,
        activation="relu"
    ),

    tf.keras.layers.Dense(
        8,
```

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```
        activation="relu"
    ),

    tf.keras.layers.Dense(
        1
    )
])

# Compile model
model.compile(
    optimizer="adam",
    loss="mse",
    metrics=["mae"]
)

# Display model architecture
print("\nTensorFlow Model Architecture:")
model.summary()

# TRAIN MODEL

history = model.fit(
    X_train_scaled,
    y_train,
    epochs=300,
    verbose=0
)

print("\nTraining Completed.")

# TEST MODEL

predictions = model.predict(
    X_test_scaled
).flatten()

# Evaluation
mae = mean_absolute_error(
    y_test,
    predictions
```

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```
)

mse = mean_squared_error(
    y_test,
    predictions
)

rmse = np.sqrt(mse)

r2 = r2_score(
    y_test,
    predictions
)

print("\nMean Absolute Error:")
print(mae)

print("\nMean Squared Error:")
print(mse)

print("\nRoot Mean Squared Error:")
print(rmse)

print("\nR2 Score:")
print(r2)

# ACTUAL VS PREDICTED

result = pd.DataFrame({
    "Actual Population": y_test.values,
    "Predicted Population": predictions
})

print("\nActual vs Predicted Population:")
print(result)

# TRAINING LOSS GRAPH

plt.figure(figsize=(8, 5))
```

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```
plt.plot(  
    history.history["loss"]  
)  
  
plt.xlabel("Epoch")  
plt.ylabel("Loss")  
plt.title("TensorFlow Training Loss")  
  
plt.grid()  
  
plt.show()  
  
# ACTUAL VS PREDICTED GRAPH  
  
plt.figure(figsize=(8, 5))  
  
plt.scatter(  
    y_test,  
    predictions  
)  
  
plt.xlabel(  
    "Actual Population (Million)"  
)  
  
plt.ylabel(  
    "Predicted Population (Million)"  
)  
  
plt.title(  
    "Actual vs Predicted Population"  
)  
  
plt.grid()  
plt.show()
```

#OUTPUT:

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colab.research.google.com/drive/1khD5QF6_G9T69OCA1hJoXqYAqduay2dr#scrollTo=j2snGGCoczkK

T105_AI_Prac10.ipynb

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Commands + Code + Text Run all

First 5 rows:

	City	Latitude	Longitude	Population_Million
0	Tokyo	35.6895	139.6917	37.4
1	Delhi	28.6139	77.2090	29.3
2	Shanghai	31.2304	121.4737	26.3
3	São Paulo	-23.5505	-46.6333	21.8
4	Mexico City	19.4326	-99.1332	21.6

Dataset Shape:
(10, 4)

Columns:
Index(['City', 'Latitude', 'Longitude', 'Population_Million'], dtype='object')

TensorFlow Model Architecture:
Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 16)	48
dense_1 (Dense)	(None, 8)	136
dense_2 (Dense)	(None, 1)	9

Total params: 193 (772.00 B)
Trainable params: 193 (772.00 B)
Non-trainable params: 0 (0.00 B)

Training Completed.
1/1 0s 75ms/step

Mean Absolute Error:
9.411681238810221

Mean Squared Error:
130.9702321923476

Root Mean Squared Error:
11.444222655661134

R2 Score:
-4.909616412970662

Actual vs Predicted Population:

	Actual Population	Predicted Population
0	18.5	17.023960
1	29.3	11.876375
2	20.4	11.064621

TensorFlow Training Loss



Variables Terminal

11:04 PM Python 3

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